

Project Title: SIEM Threat Detec on Lab Using Splunk & Atomic Red Team

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Tools: Splunk Enterprise, Splunk Universal Forwarder, Sysmon, Windows 10 VM, Ubuntu Server VM, VirtualBox

Summary:-

This project focused on building a simulated Security Operations Centre (SOC) environment to practice real-world threat detection using Splunk as a SIEM tool. A Windows 10 virtual machine was configured to generate event logs and simulate adversary behavior using Atomic Red Team, while an Ubuntu Server hosted Splunk Enterprise to ingest and analyse the data. The project emphasized detection of key attack techniques aligned with the MITRE ATT&CK framework, including suspicious PowerShell execution, brute-force login attempts, and registry-based persistence. Log forwarding was implemented using Splunk Universal Forwarder, and system-level monitoring was enhanced using Sysmon to capture detailed Windows telemetry. Multiple Splunk Reports, dashboards, and alert rules were developed to monitor these activities in real-time. Advanced SPL queries were written to detect anomalies and simulate threat hunting use cases, including correlation logic such as brute-force attacks followed by successful logins and PowerShell execution followed by registry modifications.